04-24 第一篇

The Invention of the Telescope: Seeing Beyond

Before the telescope was invented, people could only observe the sky with their eyes. The moon and stars looked small and unclear, and many secrets of the universe（宇宙）were hidden from view. [1] Around 1608, a Dutch（荷兰的） man named Hans Lippershey put two lenses（镜片） together and found that distant objects appeared much closer. Soon after, the Italian scientist Galileo heard about this invention and made his own telescope. With it, he discovered that the moon had mountains and that Jupiter（木星）had moons of its own. The telescope had changed how humans saw the heavens forever.

The basic idea of a telescope is simple: it collects and focuses light so that faraway things look larger and clearer. [2] A lens at the front gathers light from a distant object, and another lens near the eye magnifies（放大） the image. Galileo's early telescopes could make objects appear about twenty times larger. Later, scientists improved the design by using curved mirrors instead of lenses, which helped reduce distortion（失真）and allowed even bigger, more powerful telescopes to be built.

Over the centuries, telescopes have grown stronger and more advanced. Today, huge telescopes on mountains and in space can see billions of light-years away. [3] They have helped us discover new planets beyond our solar system, observe black holes, and study how galaxies（星系） form. Every few years, new space telescopes are launched, each one able to see farther and more clearly than the one before. These tools have opened our eyes to the true size and wonder of the universe.

Building and operating large telescopes is expensive and requires teamwork from many countries. [4] Engineers, scientists, and technicians from around the world work together to design, build, and maintain these instruments. Some telescopes are placed in space to avoid interference（干扰） from Earth's atmosphere. Through international cooperation, humans have achieved what no single nation could do alone—exploring the deepest reaches of space.

[5] From a simple toy in a Dutch workshop to a window on the universe, the telescope stands as one of the most important inventions in human history. It has turned our curiosity about the stars into real knowledge. As technology improves, the next generation of telescopes will surely show us even more wonders that we cannot yet imagine.

选项

A. In short, the telescope has changed not only what we see but also how we understand our place in the universe. B. The invention did not come from one person alone; several

people claimed credit at around the same time. C. The way it works depends on how light is bent and focused through lenses or mirrors. D. That is why large telescope projects often involve many nations sharing both the cost and the discoveries. E. In other words, telescopes have become our best tools for exploring the mysteries of space. F. Astronomy（天文学） has benefited greatly from these powerful instruments.

答案键（Answer Key）

[1] B [2] C [3] E [4] D [5] A

深度解析

- **[1] B**：
  - **线索**：引出望远镜发明的背景与起源。空前描述"people could only observe the sky with their eyes...secrets of the universe were hidden"，空后具体介绍"Around 1608, a Dutch man named Hans Lippershey...Galileo heard about this invention and made his own"。B选项"The invention did not come from one person alone; several people claimed credit at around the same time"（这项发明并非来自一人；当时有几人几乎同时声称自己发明）中"did not come from one person alone"自然承接空前对"发明前"的铺垫，与空后"Dutch man...Italian scientist...heard about"形成呼应——正是多人几乎同时发明，才有传播与改进，衔接顺畅。
- **[2] C**：
  - **线索**：望远镜工作原理的总领与解释。空前说明"basic idea is simple: it collects and focuses light"，空后详述"A lens at the front gathers light...another lens near the eye magnifies the image...curved mirrors instead of lenses"。C选项"The way it works depends on how light is bent and focused through lenses or mirrors"（它的工作原理取决于光如何通过镜片或镜子被弯曲和聚焦）中"the way it works"与空前"basic idea"对应，"light is bent and focused through lenses or mirrors"与空后具体描述完美衔接，逻辑紧密。
- **[3] E**：
  - **线索**：现代望远镜的成就与功能概括。空前列举"huge telescopes...can see billions of light-years away...discover new planets...observe black holes...study how galaxies form...new space telescopes are launched"，空后继续"These tools have opened our eyes to the true size and wonder"。E选项"In other words, telescopes have become our best tools for exploring the mysteries of space"（换言之，望远镜已成为我们探索太空奥秘的最佳工具）中"In other words"对上文列举的成就进行总结，"best tools for exploring the mysteries"与空后"opened our eyes to...true size and wonder"对应，衔接自然。

- [4] D：
  - **线索**：大型望远镜的建造需要国际合作。空前描述"Building and operating large telescopes is expensive and requires teamwork from many countries...Engineers, scientists...from around the world work together...Some telescopes are placed in space"，空后说明"Through international cooperation, humans have achieved what no single nation could do alone"。D选项"That is why large telescope projects often involve many nations sharing both the cost and the discoveries"（因此大型望远镜项目往往需要多国共同承担成本、分享发现）中"That is why"承接空前"expensive...requires teamwork"，"many nations sharing both the cost and the discoveries"与空后"international cooperation...no single nation could do alone"完美呼应，衔接精准。
- [5] A：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后列举"From a simple toy...to a window on the universe...turned our curiosity...into real knowledge...next generation will show us even more wonders"。A选项"In short, the telescope has changed not only what we see but also how we understand our place in the universe"（简言之，望远镜不仅改变了我们所见的，也改变了我们对自身在宇宙中位置的理解）用"In short"总结全文，"changed not only what we see but also how we understand"与空后"opened our eyes...curiosity into knowledge"形成升华，收束得当。
- **干扰项分析**：
  - **F. Astronomy has benefited greatly from these powerful instruments.**：强调天文学受益于望远镜，易被误填[3]或[5]。但F为事实陈述，缺少逻辑衔接词（如"In other words""In short"）；空[3]空前已列举具体成就，需概括性过渡句，E的"In other words...best tools"更贴切；空[5]需全文总结升华，F仅论及天文学受益，未涵盖"改变人类认知"的更广泛意义。F属"主题相关但功能与层次不足"型干扰。